

The Peacekeepers' Arms Race: Testing the Stability-Instability Paradox in the Conventional Precision-Strike Era

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Abstract

Does building better weapons make major war less likely while making smaller wars more likely? This paper tests that proposition — the stability-instability paradox — using a 192-country panel spanning 1989–2024 that triangulates nine open conflict, capability, and governance datasets from five independent providers. Three research questions are addressed. First, does higher military capability shift the composition of a country's conflict participation from interstate war toward minor and extraterritorial engagement? Two-way fixed-effects panel regression finds no such substitution: capability is associated with significantly *more* of both war and minor-conflict participation, and — most strongly — with deadlier conflicts once they occur. Second, do specific weapon-class acquisitions (aircraft, missiles, naval, ground systems) precede increases in low-intensity conflict within a short window? A panel Granger-causality test nominally flags eight of twelve pre-specified weapon-outcome pairs, but a methodological audit finds the test's standard significance procedure is itself miscalibrated for this kind of serially structured panel data; under a corrected permutation null with multiple-testing correction, zero pairs survive — a conclusion independently reproduced by an out-of-sample forecasting benchmark asking a completely different question. Third, do countries cluster into stable conflict-capability archetypes? Four data-driven archetypes emerge and predict future conflict out-of-sample, but only half are statistically stable under a correctly specified bootstrap threshold. Together the results support an “amplification, not substitution” reading of the paradox, and demonstrate — as a methodological contribution in its own right — how a widely used panel Granger-causality test can silently fail on exactly the kind of persistent, unevenly distributed treatment series common in defense-economics panels.

Keywords: stability-instability paradox, military capability, panel Granger causality, conflict composition, cluster analysis, false discovery rate

1. Introduction

Does creating superior weapons lead to peace, or does it simply change what war looks like? Deterrence theory's classical answer, formed in the shadow of nuclear weapons, is unsettling in its own way: advanced military capability can suppress the kind of war it deters most directly — large, direct, state-on-state confrontation — while doing nothing to suppress, and possibly encouraging, everything below that threshold. Glenn Snyder named this tension the stability-instability paradox in 1965: as the capability to inflict catastrophic costs rises, the *strategic* stability that capability buys at the top of the escalation ladder can coexist with, or even licence, greater *instability* lower down it — proxy wars, border skirmishes, insurgent campaigns, and other forms of conflict that stay safely under the threshold that would trigger the deterrent. The paradox has mostly been studied as a story about nuclear weapons, most influentially in the South Asian context by Kapur (2008) and in a broader empirical re-examination by Rauchhaus (2009).

This paper asks whether the same mechanism now operates one rung down the technology ladder. The twenty-first century’s most consequential capability shift has not been in nuclear arsenals — those have been comparatively static for decades — but in conventional precision-strike capability: guided munitions, combat drones, stand-off missiles, and networked targeting that let a state project lethal force with a fraction of the cost, risk, and political exposure that force projection required a generation ago. If the stability-instability logic generalises beyond the nuclear case, it should be visible here: states that accumulate this kind of capability should not simply fight more or less — they should fight *differently*, shifting the composition of their conflict participation away from costly interstate war and toward smaller, often extraterritorial, engagements where the threshold for entry has fallen. Recent examples raise the stakes of getting this right: Ukraine’s deep-strike drone campaigns against targets deep inside Russia, Iran’s drone-enabled proxy network across the Middle East, Azerbaijan’s drone-led offensive in Nagorno-Karabakh, and Turkey’s export of the Bayraktar TB2 have all been read, anecdotally, as instances of exactly this dynamic — mid-tier states projecting force in ways that used to require superpower-scale investment. Whether that anecdotal pattern holds up at panel scale, across the full universe of states and four decades of data, is an empirical question this paper is built to answer, not assume.

The project poses three linked research questions, each targeting a different layer of the same underlying claim. **RQ1** asks whether higher military capability is associated with a *compositional* shift in conflict participation — away from interstate war, toward minor and extraterritorial conflict — once country-fixed traits, economic size, regime type, and global time trends are controlled for. This is deliberately not a test of whether capability correlates with more or less conflict overall; the substitution hypothesis under test is about the *mix*, which is why the empirical strategy below treats the ratio of war to minor-conflict participation as a first-class outcome rather than an afterthought. **RQ2** narrows the question to mechanism and timing: do acquisitions of specific weapon classes — aircraft and drones, missile systems, naval platforms, ground systems — precede increases in low-intensity conflict activity within a short window, the way a “drone effect” story would predict? **RQ3** asks a structural question underneath both: do countries cluster into distinct, identifiable conflict-capability archetypes, and are those archetypes stable enough, and predictive enough, to be treated as real categories rather than statistical noise?

The headline results, previewed here so the reader knows what to expect from the rest of the paper, are not uniformly what a naive reading of the paradox would predict. RQ1 finds amplification, not substitution: higher-capability states participate in significantly more conflict of *both* types, with the cleanest and strongest signal appearing not in conflict frequency at all but in conflict *lethality* — higher-capability states fight measurably deadlier conflicts. RQ2’s contribution turns out to be as much methodological as substantive, and this is worth flagging plainly before Section 5 rather than leaving it as a surprise: the standard significance test conventionally used for this class of panel Granger-causality problem is shown to be miscalibrated on exactly this kind of serially persistent, unevenly distributed treatment data, and once that miscalibration is corrected for, the lead-lag signal that looked promising under the naive test disappears entirely — a conclusion independently corroborated by a genuinely out-of-sample forecasting exercise that never touches a p-value at all. RQ3 finds four data-driven archetypes that carry real predictive information about a country’s future conflict trajectory, but only half of them meet a correctly calibrated bar for statistical stability, a result presented here as a genuine mixed finding rather than smoothed into either “the clusters are real” or “the clusters are noise.”

The remainder of the paper proceeds as follows. Section 2 sets out the theoretical framework and derives formal hypotheses for each research question. Section 3 describes the data. Section 4 details

the empirical strategy for all three research questions, including the methodological correction that RQ2 required. Section 5 reports results. Section 6 discusses what the findings mean for the stability-instability paradox and for applied panel-causality practice more broadly. Sections 7 and 8 cover limitations and conclude.

2. Theoretical Framework and Hypotheses

2.1 The stability-instability paradox

The stability-instability paradox describes a specific, non-obvious relationship between a state's capacity to deter catastrophic war and its willingness to tolerate conflict below that threshold. The term originates with Snyder (1965), who argued that as mutual deterrence capability rises between two actors, the resulting stability at the highest level of potential conflict — neither side rationally initiates a war that both know would be catastrophic — does not translate into stability across the board. Instead, that same high-level deterrence can create room for lower-intensity confrontation, because both sides understand that neither has an incentive to escalate a small conflict into the kind of confrontation that deterrence exists to prevent. The logic is genuinely paradoxical rather than merely counterintuitive: it is not that deterrence *fails* to work at the top of the ladder — by hypothesis it works very well there — but that its success at the top is precisely what removes the main check on behaviour lower down. A state confident that a rival will not escalate a small confrontation into a large one has, in effect, been handed more room to manoeuvre below the threshold that the rival's deterrent actually covers, not less.

Robert Jervis developed a closely related and often-paired argument about the security dilemma and the paradoxical effects of defensive capability more generally (Jervis, 1984). Where Snyder's version of the paradox is framed around mutual deterrence between two roughly matched actors, the Jervis-style security-dilemma framing generalises the underlying mechanism: capability that a state acquires for defensive or deterrent purposes can be read by others as offensive-capable, and can simultaneously alter the acquiring state's own calculus about what level of risk-taking is now tolerable. Both lines of argument converge on the same structural claim that motivates this project — that capability's stabilising and destabilising effects are not independent of each other, they are the *same* mechanism observed at two different rungs of the escalation ladder. Together, Snyder's and Jervis's arguments form the theoretical core of the paradox: capability that stabilises the top of the escalation ladder can destabilise everything beneath it.

The paradox was developed initially in the context of superpower nuclear deterrence, but its most influential recent empirical tests have come from scholars asking whether it generalises to regional nuclear dyads. Kapur (2008) examines the paradox in South Asia, arguing that the region's stability at the nuclear level has coincided with — and plausibly enabled — a decade of sustained sub-conventional conflict between India and Pakistan. Rauchhaus (2009) undertakes a broader, more systematic empirical re-examination of the paradox across the population of nuclear dyads, providing quantitative grounding for a claim that had previously rested mostly on case-study reasoning. Both studies treat the paradox as operating specifically through nuclear weapons: the capability doing the stabilising work is the capacity for mutually assured destruction, and the instability it enables is conventional or sub-conventional conflict that stays below the nuclear threshold.

2.2 From nuclear to conventional: the twenty-first-century question

This project’s motivating claim is that the same underlying logic — extreme capability at the top of the escalation ladder licensing more permissive behaviour beneath it — may no longer be a story that is exclusively about nuclear weapons. Conventional precision-strike capability has, over the past two decades, acquired some of the same escalation-dampening properties that nuclear weapons have long held at a much higher level of destructive potential: it lets a state impose costs, gather intelligence, and coerce an adversary from a position that carries comparatively little risk of retaliation against the attacker’s own territory or forces. If that shift in the *cost structure* of force projection is doing similar theoretical work to what nuclear deterrence does in Kapur’s and Rauchhaus’s accounts, it should generate an analogous compositional signature: states that accumulate more conventional military technology should not simply become more or less conflict-prone in an undifferentiated sense, but should shift the *mix* of conflict they participate in — away from the kind of interstate war that advanced capability is best at deterring, and toward the low-intensity, frequently extraterritorial engagements (drone strikes, proxy support, cross-border raids) that advanced capability makes cheaper to initiate.

The plausibility of this extension rests on a specific empirical premise worth stating explicitly: that the marginal cost of *initiating* a small, discrete use of force has fallen sharply for states that hold precision-strike, unmanned, and networked-targeting capability, independent of whatever is happening at the level of large-scale deterrence. A loitering munition or an armed drone lets a state strike a target without risking a pilot, without the diplomatic exposure of a manned incursion, and often without publicly claiming responsibility at all — the per-engagement political and material cost of a small strike has fallen even where the strategic logic deterring a full-scale war between the same two states has not changed at all. Recent, widely observed cases are consistent with exactly this reading: Ukraine’s deep-strike drone campaigns against targets inside Russian territory, Iran’s drone-enabled proxy operations across the Middle East, Azerbaijan’s drone-led offensive in the 2020 Nagorno-Karabakh war, and Turkey’s export of the Bayraktar TB2 to multiple mid-tier militaries have each been read, in commentary and area-studies work, as instances of a state or non-state actor using precision unmanned systems to project force in ways that previously required superpower-scale investment. Those examples are illustrative rather than probative on their own — a handful of high-profile cases cannot establish a general pattern — which is precisely why RQ2 is designed to test whether the same pattern is visible systematically across the full panel of states, rather than resting the claim on anecdote.

2.3 The substitution hypothesis, stated precisely

A common misreading of the stability-instability paradox treats it as a claim that advanced weapons cause more conflict overall. That is not the claim under test here. The correct reading, and the one this project is built to evaluate, is *compositional*: advanced weapons change the *kind* of conflict a state engages in, not necessarily the total amount. This is why the primary dependent variable of interest for RQ1 is not a count of conflicts of any single type, but the *ratio* of war-level to minor-level conflict participation — a variable constructed specifically because it is the quantity a genuine substitution effect would move, and a quantity that a simple “more capability, more conflict” story would not move in any particular direction on its own.

2.4 Formal hypotheses

RQ1 — Compositional shift. If the substitution mechanism operates as theorised, higher military capability should be associated with: a lower rate of interstate war participation (H1a: $\beta < 0$ for war-level conflict participation); a higher rate of minor-conflict participation (H1b: $\beta > 0$ for minor-conflict

participation); a lower war-to-minor participation ratio (H1c: $\beta < 0$ — the direct test of the compositional claim); a higher rate of extraterritorial conflict participation (H1d: $\beta > 0$); and a higher share of a country’s conflict activity that is extraterritorial rather than domestic (H1e: $\beta > 0$). All five are evaluated jointly; a genuine substitution finding requires the pattern to hold across the set, not just in one outcome considered in isolation.

RQ2 — Weapon-class lead-lag. If specific weapon-class acquisitions are a proximate mechanism through which the paradox operates at the conventional level — the “drone effect” story in its most literal, testable form — then increases in a country’s weapon-class deliveries (aircraft and drones, missiles, naval platforms, ground systems) should systematically precede increases in that country’s low-intensity conflict activity within a short window, more often and more strongly than chance would predict, across the panel of states (H2).

RQ3 — Country archetypes. If capability and conflict behaviour co-vary in the patterned way the paradox implies, states should not be scattered uniformly across capability-conflict space but should cluster into a small number of distinguishable archetypes — provisionally, something like high-capability states that mostly avoid war (“Safe Hegemons”), high-capability states that sustain elevated low-intensity conflict (“Armed Instabilizers”), lower-capability states that nonetheless deter effectively (“Defensive Deterrents”), and states largely disengaged from organised conflict altogether — and those clusters should be both statistically stable under resampling and informative about states’ future conflict trajectories (H3).

Each hypothesis is stated so that it can fail cleanly rather than be argued away after the fact. H1a–H1e are jointly falsified, in the sense relevant to the substitution claim, if the war-to-minor ratio (H1c) shows no significant relationship to capability regardless of what happens to the count-based outcomes individually — a compositional claim that is not rescued by a significant result on any single count variable considered alone. H2 is falsified if the corrected, appropriately calibrated significance test finds no weapon-class cell where deliveries precede conflict-participation change more often than a null with the same serial structure would produce by chance. H3 is falsified, or at minimum substantially qualified, if the clusters recovered are not both stable under resampling and predictive out of sample; a partition that is stable but uninformative, or informative but statistically indistinguishable from a resampling artefact, would not support treating the archetypes as genuine structural categories.

3. Data and Measures

3.1 Sources

The analysis triangulates nine open, peer-reviewed datasets from five independent providers (SIPRI, UCDP, V-Dem, the World Bank, and the Correlates of War project), each keyed to ISO 3166-1 alpha-3 country codes to support cross-source merging.

Dataset	Role in analysis
SIPRI Military Expenditure Database (MILEX)	Annual capability proxy; a standard measure in Correlates of War (COW), PRIO, and RAND quantitative international-relations research.
SIPRI Arms Transfers Database (Trend Indicator Value, TIV)	Year-of-delivery arms-transfer data disaggregated by weapon category; central to RQ2’s weapon-class lead-lag test.

Dataset	Role in analysis
UCDP/PRIO Armed Conflict Dataset (ACD)	Conflict type (interstate / intrastate / extrasystemic) and intensity (minor / war); the primary conflict-type source for RQ1.
UCDP Georeferenced Event Dataset (GED)	Individual dated conflict events, enabling the finer time resolution RQ2's lead-lag test requires.
UCDP Dyadic Dataset	Participant-level (side A / side B) conflict data, used to build a <i>participation</i> panel — who is fighting, not merely where the fighting occurs — which corrects a systematic attribution problem for extraterritorial conflicts discussed below.
UCDP Battle-Related Deaths (BRD)	Per-conflict fatality estimates, used to construct the battle-death intensity measure reported in Section 5.1.
V-Dem (Varieties of Democracy)	Regime-type and democracy scores, included as a confounder control since democracies and autocracies exhibit systematically different conflict patterns independent of capability.
World Bank World Development Indicators (WDI)	GDP, GDP per capita, and population, used as controls for country economic size.
Correlates of War (CoW) National Material Capabilities (CINC) and Formal Alliances	CINC serves as an independent benchmark against which the capability measure constructed here is validated, not as a substitute for it; Formal Alliances data support alliance-structure robustness checks.

The Uppsala Conflict Data Program (UCDP) supplies four of the nine datasets in the table above and is worth a brief note on its own authority as a source, since it is doing the most conceptual work of any single provider in this analysis. UCDP is widely regarded as the leading source of data on organised armed violence internationally, and its operational definitions of what counts as an armed conflict — a threshold built around sustained, organised violence rather than isolated incidents — have become close to a global standard for how conflicts are systematically categorised in quantitative work. Its data collection began in 1979 and was formalised in the mid-1980s, making it the longest continuously maintained effort of its kind; that continuity is part of why UCDP, rather than a shorter-running or more narrowly scoped alternative, was chosen as the conflict-side backbone of this project's outcome variables.

3.2 Military Technology Score (MTS)

Rather than commit to a single capability proxy, three independent specifications are constructed and reported in parallel: a MILEX-only specification (log-normalised mean annual military expenditure in constant-dollar terms), a TIV-only specification (log-normalised cumulative SIPRI Trend Indicator Value across weapon categories), and a PCA composite specification (the first principal component of MILEX, TIV, and selected CINC subcomponents, which lets the data determine relative weighting rather than imposing one). All three are carried through every RQ1 regression. A finding that holds across all three specifications is reported as robust; a finding that depends on which single-indicator specification is used is flagged as specification-dependent rather than treated as general.

The three specifications are deliberately not redundant with one another. MILEX captures total defence spending, which reflects manpower, maintenance, and infrastructure as much as it reflects frontier weapons technology; TIV captures the flow of specific weapon systems into a country’s arsenal, which is closer to a direct measure of technological capability but says nothing about domestic production or spending on personnel; and the PCA composite, by construction, weights whichever of the underlying indicators — including relevant CINC subcomponents — actually co-varies most with the others in the observed data, rather than imposing a single analyst’s judgement about which indicator is “correct.” Reporting all three in parallel, rather than picking one as primary and mentioning the others only as an appendix check, is the mechanism by which this paper avoids overstating the generality of any single result — a discipline that matters most, and pays off most directly, in Section 5.1, where two of the paper’s core findings turn out to be significant under some specifications and not others.

3.3 Coverage

The merged, ISO3-standardised analysis panel covers 192 countries. While the underlying conflict data (UCDP’s Armed Conflict Dataset) nominally extends back to 1946, the effective analysis window used throughout this paper is 1989–2024: UCDP’s Georeferenced Event Dataset and Battle-Related Deaths series both begin in 1989, V-Dem’s country-year coverage has substantial gaps before that point, and SIPRI’s deal-level TIV data are considerably less complete for the Cold War’s earlier decades. Binding the analysis to the post-Cold War window keeps every outcome and every MTS specification jointly estimable on the same sample rather than mixing eras with very different data completeness. Within this window the merged panel comprises 6,912 country-year observations; the primary regression sample after listwise deletion for missing controls is 4,896 country-years for the count-based outcomes and 1,846 for the two ratio-based outcomes (war-to-minor ratio and extraterritorial share), which are undefined for country-years with no recorded conflict activity. RQ3’s clustering sample is smaller still: 157 countries with sufficient feature coverage across four ten-year epochs (1990s–2020s), yielding 610 country-epoch observations — the unit of analysis for the archetype clustering reported in Section 5.3.

Conflict participation is built as a *participant*-level panel from UCDP’s Dyadic Dataset (which country side was fighting) rather than a *location*-level panel derived from where the fighting physically occurred. This distinction matters specifically for the extraterritorial-conflict outcomes central to RQ1: a location-based approach would attribute a conflict entirely to the country on whose soil it took place, which systematically misses external participants — coalition members in a multi-state intervention, or a state materially supporting one side of a conflict fought on someone else’s territory — and would understate exactly the kind of externally projected, low-footprint engagement the paradox’s instability channel is theorised to produce. Building the panel around participation rather than location is one of the design choices, alongside the six pipeline corrections summarised below, that this project treats as part of the evidence for taking its results seriously rather than as an incidental implementation detail.

3.4 Data-quality assurance

Six data-quality issues were identified and corrected during pipeline construction; each is noted briefly here because catching them is itself part of the evidence for the reliability of the results that follow, not because the details matter to the substantive argument. A fuzzy country-name matcher initially resolved “Islamic State” to Iceland (ISO3 = ISL), inflating Iceland’s apparent conflict count; this was caught by a country-resolution spot-check and required a dedicated resolver with per-dataset override rules rather than reliance on any single library’s matching logic. SIPRI’s arms-transfer query portal was found, via manual audit against SIPRI’s own published totals, to silently return export rather

than import totals regardless of the user-interface selection, which would have inverted the capability proxy for every exporting country if left uncaught. UCDP’s dyadic data fold multi-state coalitions into a single location code, which was found — via a spot-check on the 2003 Iraq War — to systematically undercount coalition members as extraterritorial participants, exactly the outcome RQ1 is testing. Historical-state variables (e.g., for the USSR, Yugoslavia, and Czechoslovakia) were initially aggregated across successor states using an arithmetic mean rather than a sum, silently halving expenditure and population figures for any country touched by one of those splits. The TIV series used a literal zero for both “no data reported” and “genuinely zero imports,” biasing roughly 5,400 country-years toward the bottom of the capability scale until the two were disambiguated. Finally, an internal pipeline check found that a downstream robustness flag in the original RQ2 analysis had been computed in memory after the corresponding results table was already written to disk, meaning the saved file never reflected the flag at all — caught only by a later notebook’s attempt to re-read the saved file, illustrating why the results reported below are cross-checked against the actual saved output of each analysis stage rather than trusted from memory.

4. Empirical Strategy

4.1 RQ1: two-way fixed-effects panel regression

RQ1 is estimated with a two-way fixed-effects panel regression of the form

$$\text{conflict_outcome}_{i,t} = \beta \cdot \text{MTS}_{i,t} + X'_{i,t}\delta + \alpha_i + \gamma_t + \varepsilon_{i,t}$$

where α_i is a country fixed effect absorbing all time-invariant country traits, γ_t is a year fixed effect absorbing global time trends common to all countries, and $X_{i,t}$ is a vector of time-varying controls (log GDP, log population, V-Dem regime score). Standard errors are clustered by country throughout. The model is estimated separately for each of five conflict-composition outcomes (war participation, minor-conflict participation, the war-to-minor ratio, extraterritorial participation, and extraterritorial share) and for battle-death intensity, under each of the three MTS specifications, with a Poisson fixed-effects specification and a set of Cold War / post-Cold War / post-9/11 subperiod splits run as robustness checks on the count-based outcomes.

4.2 RQ2: from cross-correlation to panel Granger causality — and why the standard test needed correcting

RQ2 begins with SIPRI TIV annual deliveries, disaggregated into four weapon classes (aircraft and unmanned systems; missiles and precision munitions; naval platforms; ground systems), as the treatment series, and UCDP GED-derived conflict-participation counts as the outcome series. A first-differenced, per-country cross-correlation function is computed at lags of 0–36 months as a purely descriptive view of lead-lag structure. The inferential test is the Dumitrescu and Hurlin (2012) panel Granger-causality test, which asks whether, averaged across the panel of states, a country’s own past weapon-class deliveries carry information about that country’s future conflict activity beyond what the conflict series’ own past already provides — a test of *predictive precedence*, explicitly not of causation.

Applied in its textbook form, the Dumitrescu-Hurlin test’s significance is assessed against a within-country shuffle null: each country’s treatment series is randomly permuted in time, breaking any tem-

poral alignment with the outcome, and the test statistic’s distribution under many such permutations defines the null. This is the standard approach in applied panel Granger-causality work. It turns out to be the wrong null for weapon-transfer panel data specifically, because a random within-country shuffle destroys a feature of the treatment series that has nothing to do with the outcome at all — its own negative serial (autocorrelation) structure, a pattern in year-to-year arms deliveries that a random shuffle erases but that a real trajectory of deliveries genuinely has. Because the test’s significance calibration implicitly assumes the null series looks like the real series in every respect except its relationship to the outcome, destroying that serial structure makes the shuffle-based null too “well-behaved” relative to the real, more erratic treatment series, and the test over-rejects as a direct consequence — it flags relationships as significant more often than its own nominal error rate promises, on data with this specific structure, regardless of whether a real lead-lag relationship exists. The correction applied here is a *circular-shift* null: instead of randomly permuting each country’s treatment values, each country’s series is rotated by a random offset, which preserves its own autocorrelation structure exactly while still destroying its temporal alignment with the outcome. Because circular shifting only changes *when* a treatment shock happens, not the shape of the series that experiences it, it isolates the null hypothesis genuinely under test — no relationship between treatment and outcome — without also changing something else about the data that the naive shuffle changes as a side effect. Statistical significance under the corrected null is then assessed per weapon-outcome cell at lag 1 (the only lag at which the raw test produces positive signal at all; lags of 2–3 years are shown in Section 5.2 to be uninformative), with Benjamini and Hochberg’s (1995) false discovery rate procedure applied across the twelve pre-specified weapon-by-outcome cells to control for testing multiple hypotheses at once.

A second, independent test is run alongside the corrected inferential test: a genuinely out-of-sample forecasting benchmark. Rather than asking whether a lead-lag relationship is statistically distinguishable from noise, this benchmark asks a decision-relevant question directly — does knowing a country’s weapon-class acquisitions actually improve real forecasts of that country’s next-year conflict-participation change, over and above what a simple autoregressive baseline already achieves? A rolling-origin design is used: at each yearly forecast origin, models are trained only on data available up to that point and evaluated on the following year, exactly as a real forecaster would be constrained. The treatment-augmented model’s forecast error is compared against an AR(1)-only baseline that uses each country’s own most recent conflict level and nothing else, and the resulting skill differential is itself tested against the same circular-shift permutation null used for the inferential test, so that the two independent approaches — inference and forecasting — are held to the same standard of statistical rigor.

4.3 RQ3: clustering with stability and out-of-sample validation

RQ3 clusters countries on a feature set that includes each MTS specification, conflict counts and ratios by type, extraterritorial-involvement rate, casualty intensity, V-Dem regime score, and region. Both K-Means and Ward hierarchical clustering are run for k from 2 to 10 (this paper reports the tested range 2–8, at which the highest silhouette score is attained), with the number of clusters selected via silhouette score and the gap statistic rather than fixed in advance. Cluster stability is assessed by bootstrap-resampling the country sample 100 times and computing the Jaccard similarity of cluster assignments across resamples. The conventional stability threshold used for this diagnostic in much applied cluster-analysis work — a mean Jaccard coefficient above 0.75 — is calibrated for cross-validation-style resampling schemes, where a fresh, non-overlapping subsample is drawn each time. Bootstrap-with-replacement resampling is a different scheme: because a bootstrap sample of size n necessarily duplicates roughly a third of the original observations and omits roughly a third, cluster

assignments recovered under it are mechanically noisier than under cross-validation resampling even when the underlying partition is genuinely stable, and Ben-Hur, Elisseeff, and Guyon (2002) show that the appropriate stability ceiling for bootstrap-with-replacement schemes at $n=157$, the sample size of this project's RQ3 clustering, is closer to 0.50 than to 0.75. This paper therefore reports cluster stability against the theoretically appropriate 0.50 threshold rather than the 0.75 convention, and states this reasoning here so that Section 5.3 can report the resulting stability split without re-deriving the justification from scratch. Independently of stability, out-of-sample predictive validity is assessed by holding out 2019–2024 conflict data entirely: clusters are trained on pre-2019 features only, and a one-way ANOVA tests whether the resulting cluster membership predicts each country's post-2019 conflict activity.

5. Results

5.1 RQ1 — Compositional Shift

The central finding of RQ1 is that higher military capability is not associated with a compositional shift away from war and toward minor conflict; it is associated with significantly more of both, and, most strongly, with more lethal conflict once it occurs. Under the primary PCA-composite MTS specification, war-level conflict participation rises significantly with capability ($\beta = 0.30$, $SE = 0.14$, $p = .037$, $N = 4,896$) — the wrong sign relative to the deterrence-consistent prediction that higher capability should suppress war participation (H1a predicted $\beta < 0$). Minor-conflict participation also rises significantly with capability ($\beta = 0.54$, $SE = 0.17$, $p = .002$, $N = 4,896$), the direction H1b predicted, but the fact that *both* outcomes move in the same direction rather than trading off against each other is itself the substantive finding: it is amplification, not substitution. The most direct test of the substitution claim, the war-to-minor participation ratio, shows no significant relationship at all ($\beta = 0.23$, $SE = 0.22$, $p = .294$, $N = 1,846$) — and its point estimate is positive, the opposite sign from what a genuine substitution effect (H1c: $\beta < 0$) would produce. Extraterritorial participation and extraterritorial share are both statistically null in the full sample ($\beta = 0.16$, $p = .427$ and $\beta = -0.51$, $p = .124$, respectively; H1d and H1e predicted positive coefficients for both).

The single strongest result in the entire project is not one of the five compositional outcomes at all: battle-death intensity (log-transformed battle-related deaths) rises sharply and significantly with capability ($\beta = 2.03$, $SE = 0.59$, $p = .001$, $N = 4,896$, $R^2_{within} = .44$) under the primary specification. This finding is not driven by a handful of high-capability outlier states: a leave-one-out jackknife that re-estimates the same regression 193 times, once with each country individually removed from the sample, finds the coefficient stays positive and statistically significant in every single case, ranging narrowly between 1.82 and 2.16 across all single-country drops.

Robustness across the three independent MTS specifications is mixed in an informative way rather than uniform. Minor-conflict participation's positive association with capability is significant under the PCA-composite ($p = .002$) and MILEX-only ($p = .017$) specifications and reaches only the 10 percent level under the TIV-only specification ($p = .091$); it is also significant under a Poisson fixed-effects specification that treats the outcome as a count rather than a continuous variable ($p = .048$). War participation's positive, deterrence-opposing sign holds under all three MTS specifications and under Poisson, but only reaches conventional significance under the PCA-composite specification ($p = .037$); the MILEX-only ($p = .851$) and TIV-only ($p = .127$) point estimates are positive but not statistically distinguishable from zero. Battle-death intensity's positive coefficient is significant under the PCA-composite ($p = .001$) and TIV-only ($p = .014$) specifications, but not under the MILEX-only specification

($p = .316$) — a reminder that “robust across every specification” is a claim this paper does not make for any single outcome, even its strongest one, and reports accordingly.

A subperiod split by era offers a suggestive, though not statistically decisive, additional pattern. Extraterritorial participation’s association with capability is negative and marginally significant in the pre-9/11 window ($\beta = -0.43$, $SE = 0.25$, $p = .089$, $N = 1,572$, years 1989–2001) and flips to positive, though not significant, in the post-9/11 window ($\beta = 0.38$, $SE = 0.46$, $p = .409$, $N = 3,324$, years 2002–2024). Neither estimate individually clears the conventional significance threshold, so this is reported here as a directional pattern worth flagging for future work rather than a confirmed era-specific reversal.

Two further mechanism checks probe whether the battle-death intensity finding is a diffuse, panel-wide effect or is being driven by a specific regional or institutional subgroup, and both point toward a broad-based effect. A regional decomposition finds the battle-death intensity association is significantly concentrated in Africa (marginal $\beta = 4.95$, $p < .001$) relative to every other region tested, none of which reaches significance individually; a leave-one-out jackknife within that African subsample, dropping each of the 34 in-sample African countries one at a time plus a joint three-country drop, leaves the coefficient positive and significant throughout (β ranging from 4.49 to 5.41 across all drops), indicating the regional concentration itself is not an artefact of one or two influential countries. A separate interaction test finds no evidence that a country’s regime type moderates the capability–battle-death relationship: neither a continuous V-Dem polyarchy interaction ($p = .497$) nor a democracy/autocracy dummy interaction ($p = .687$) is significant, and the marginal effect of capability on battle-death intensity remains significant and of similar magnitude whether evaluated at low, medium, or high polyarchy scores. Read together, the battle-death intensity result looks like a broad, regionally concentrated but not outlier-driven or regime-contingent pattern, rather than an artefact of any single country, region-selection choice, or type of government.

5.2 RQ2 — Weapon-Class Lead-Lag

This section is the project’s central methodological contribution, and it is structured as a narrative arc rather than a list of statistics because the sequence in which each stage revises the previous one is the point.

(a) The original test. Run at its most literal — a one-year lag between weapon-class deliveries and conflict-participation change, the pre-specified twelve-cell grid of four weapon classes by three conflict outcomes — the Dumitrescu-Hurlin panel Granger-causality test nominally flags eight of the twelve cells as significant at conventional levels under its standard asymptotic significance procedure. The strongest apparent signal is ground-systems deliveries preceding war-level conflict participation ($Z = 6.42$, $p < .001$), followed by ground-systems deliveries preceding extraterritorial participation ($Z = 4.93$, $p < .001$) and missile deliveries preceding minor-conflict participation ($Z = 3.64$, $p < .001$). Read at face value, this is a striking result, and read at face value it is also the wrong conclusion to draw, for the reason set out next.

Even before the deeper diagnosis described below, an earlier and cruder robustness check on this same set of results had already raised a flag worth noting for the historical record. A placebo comparison, run alongside the original test, re-estimates each cell’s test statistic on placebo (randomly reassigned) treatment-outcome pairings and records how often the placebo version alone reaches significance; at lag 1, that placebo significance rate ranges from 16 percent to 50 percent across the twelve cells — three to ten times higher than the 5 percent rate a well-calibrated test should produce on data with no real relationship. That placebo check could not, on its own, distinguish a genuinely miscalibrated null

from ordinary sampling variation across a small set of placebo draws, so it did not settle the question by itself, but in hindsight it was already pointing at exactly the problem the circular-shift diagnosis below confirms directly.

(b) The discovery. The eight nominally significant cells are all built on a significance procedure calibrated against a within-country shuffle null, and that null turns out to be badly mismatched to this specific kind of data. The treatment series — a country’s own year-to-year weapon-class deliveries — has a genuine negative lag-1 autocorrelation of -0.33 in the observed data: a year of unusually large deliveries in one weapon class tends to be followed by a leaner year, a pattern plausible on its own terms (arms transfers arrive in discrete tranches tied to procurement and delivery cycles, rather than as a smooth annual flow) and, more importantly here, a genuine feature of the actual data rather than an artefact of how it was constructed. A random within-country shuffle of that same series destroys almost all of that structure, reducing the shuffled series’ own autocorrelation to -0.02 — indistinguishable from a genuinely random walk. Because the significance test’s null distribution is built entirely from data whose serial structure no longer resembles the real treatment series it is meant to stand in for, the null is too tame relative to reality, and the test rejects the no-relationship null more often than its own nominal 5 percent rate promises — not because a real relationship exists, but because the comparison itself is unfair to the null hypothesis. This is a general property of the test’s calibration on this class of data, not a quirk of one or two cells: it applies to the null distribution itself, before any particular weapon-outcome pairing is even considered.

(c) The correction. A circular-shift null is substituted for the shuffle: rather than randomly reordering each country’s treatment values, each series is rotated by a random temporal offset, which exactly preserves the series’ own autocorrelation (the rotated series retains a lag-1 autocorrelation of -0.31 , essentially matching the observed -0.33) while still fully destroying any genuine temporal alignment with the outcome. The practical consequence of using the correct null rather than the shuffle is large and can be quantified directly: rerunning the *same, unchanged* significance test 2,000 times against circular-shift-permuted data — where by construction there is no relationship between treatment and outcome — the test still rejects the null at conventional significance somewhere between 32 percent and 48 percent of the time across the twelve cells, six to ten times its own nominal 5 percent rate. That figure is the calibrated size of the naive test on this exact data structure: not an estimate from a different dataset or a simulation study in the abstract, but the same instrument, the same panel, with only the null-generating mechanism corrected to actually represent “no relationship.”

Applying Benjamini-Hochberg false discovery rate correction across the twelve pre-specified cells against this corrected null produces a starkly different picture from the naive result: zero of twelve cells survive. The cell with the strongest raw signal under the naive test, ground-systems deliveries preceding war participation, has a per-cell permutation p-value of .032 under the corrected null — nominally still below .05 in isolation — but a Benjamini-Hochberg-adjusted q-value of .390 once the full twelve-cell family is accounted for, far short of the conventional .05 bar. No other cell comes closer.

(d) Independent corroboration. A completely different question, asked with a completely different method, reaches the same answer. Rather than testing whether a lead-lag relationship is statistically distinguishable from noise, an out-of-sample rolling-origin forecasting benchmark asks whether knowing a country’s weapon-class deliveries actually improves real forecasts of next-year conflict-participation change over a simple autoregressive baseline. At each yearly forecast origin between 2005 and 2023, four models are fit on data available only up to that point and evaluated against the following year’s actual outcome: a naive-zero forecast, a historical-mean forecast, an AR(1)-only model using each country’s own most recent conflict level, and a treatment-augmented model that adds the

relevant weapon-class delivery series to the same AR(1) specification. The comparison that matters for RQ2 is the treatment-augmented model against the AR(1)-only baseline, since the AR(1) baseline already has access to everything the outcome’s own recent history can offer; any genuine value from knowing weapon-class deliveries has to show up as an improvement over that specific baseline, not merely over a naive constant.

It does not show up. The forecast-skill differential between the treatment-augmented model and the AR(1)-only baseline is close to zero and inconsistently signed across all twelve cells, ranging from a 0.02 percent forecast-error improvement (ground systems, war participation) to a 0.06 percent *degradation* (aircraft, minor conflict) relative to the baseline — differences with no practical or forecasting-relevant magnitude in either direction, and smaller in absolute terms than typical year-to-year noise in the underlying conflict series. Testing this skill differential against the same circular-shift permutation null used for the inferential test, again with Benjamini-Hochberg correction across the twelve-cell family, again produces zero survivors. This corroboration matters specifically because it is not a second run of the same test: prediction and inference are different statistical objects, entirely different code, and a different loss function, and their independent agreement is stronger evidence of a genuine null than either result would be alone — a lead-lag relationship real enough to matter substantively should have shown up in at least one of the two approaches, and it does not show up in either.

(e) The naive-versus-rigorous framing, stated plainly. Considered as a single, self-contained result, the contrast between what a naive analysis of this panel would report and what a rigorous one actually supports is itself the section’s headline finding, not a caveat attached to a null result. The naive, uncorrected Dumitrescu-Hurlin test flags 8 of 12 pre-specified weapon-outcome cells as significant. The same test, run against a null that is actually appropriate for this data’s serial structure and corrected for testing twelve hypotheses at once, finds 0 of 12 survive — a conclusion an entirely independent forecasting exercise reaches by an unrelated route. There is no robust evidence in this panel, under any of three independent tests, that weapon-class acquisitions precede shifts in low-intensity conflict within a one-to-three-year window.

5.3 RQ3 — Country Archetypes

K-Means clustering of 610 country-epoch observations (157 countries across four decade-epochs) identifies four archetypes, matching the number the project’s theoretical framing anticipated: **Safe Hegemon** (118 country-epoch assignments; MTS composite mean 0.28, low war-to-minor ratio at 0.20, and by far the highest mean battle-death intensity of any archetype at 5.90 on the log scale), **Armed Instabilizer** (125 country-epoch assignments; the highest capability of any archetype at 0.30, elevated minor-conflict participation at 1.97, and the highest extraterritorial-share mean of any archetype at 1.29), **Defensive Deterrent** (85 country-epoch assignments; moderate capability at 0.26, comparatively low overall conflict activity), and **Inert Non-Combatant** (282 country-epoch assignments, the largest group; the lowest capability of any archetype at 0.18 and minimal conflict engagement of any kind, active in only 10 percent of country-years). Most countries are not fixed to a single archetype over time: 100 of 157 countries (64%) appear in more than one archetype across the four epochs studied — direct, epoch-level evidence for the “soft regions on a continuum, not fixed types” reading argued in Section 6, rather than an assumption stated there without support in Results.

The Safe Hegemon profile is worth pausing on because it is not the uniformly quiescent category its provisional label implies. Safe Hegemon states have the lowest war-to-minor participation ratio of any archetype — consistent with the “safe” half of the label — but simultaneously the highest mean battle-death intensity of any archetype, by a wide margin over the next-highest group. That combination is

consistent with a profile of states that rarely enter war-level conflict, but whose conflicts, when they do occur, are historically among the most costly in the panel — a pattern that echoes RQ1’s own headline finding that capability’s clearest empirical signature is intensity rather than frequency, recovered here independently from an entirely different, unsupervised method.

The choice of four clusters is not the unambiguous output of a single criterion, and this project reports that disagreement rather than hiding it. The silhouette score, a measure of how well-separated clusters are from one another, peaks at $k = 4$ (silhouette = 0.293) across the tested range of $k = 2$ to 8. The gap statistic, a different and equally standard criterion for choosing the number of clusters, instead favors $k = 8$, the upper end of the tested range (gap = 1.136, still rising at the boundary of what was tested). A second, independent clustering algorithm — Ward hierarchical clustering, run on the identical feature set — shows the same pattern in its own right: Ward’s own silhouette score is also maximized at $k = 8$ (0.243) rather than $k = 4$ (0.181). Two standard, textbook criteria for choosing cluster count, applied to the same data, therefore disagree by a factor of two; $k = 4$ is used as the primary specification here because silhouette is the more robust criterion for the non-uniform, regionally clustered distribution typical of country panels, but the honest description of how $k = 4$ was chosen is “one criterion, with a documented and substantively-sized disagreement from another,” not “the data unambiguously says four.”

Bootstrap stability, assessed by resampling the country sample 100 times and computing the Jaccard similarity of cluster assignments, produces a genuinely mixed result once judged against the theoretically appropriate threshold for bootstrap-with-replacement resampling (0.50, per the reasoning in Section 4.3, rather than the 0.75 threshold conventional for cross-validation-style resampling). Two of the four archetypes clear the corrected bar: Safe Hegemon (mean Jaccard = 0.58) and Inert Non-Combatant (mean Jaccard = 0.54). Two do not: Defensive Deterrent (mean Jaccard = 0.46) and Armed Instabilizer (mean Jaccard = 0.42). This is reported as a genuine 2-of-4 split, not rounded in either direction — the clusters are neither uniformly stable nor uniformly noise.

Stability and predictive validity, however, answer different questions, and the two need not move together. Held-out, out-of-sample validation — clusters trained on pre-2019 data only, tested against whether membership predicts 2019–2024 conflict activity — finds cluster membership strongly predicts future conflict (one-way ANOVA, $F = 33.6$, $p < .001$, across four groups). All four archetypes carry genuine forward-looking information about a country’s conflict trajectory, including the two whose bootstrap stability falls short of the corrected threshold; this project treats stability and out-of-sample predictive validity as two different, non-substitutable tests of whether an unsupervised partition is “real,” and reports both rather than letting one stand in for the other.

6. Discussion

Taken together, the three results describe a paradox that operates, but not in the compositional form the classical formulation predicts. RQ1’s finding of amplification rather than substitution means that higher-capability states are not trading war for minor conflict, they are participating in more of both — which is a different theoretical claim than Snyder’s original formulation, closer to a story about capability lowering the threshold for entering *any* conflict rather than one about capability specifically redirecting states away from the costliest kind. The strongest single result in the paper, battle-death intensity rising sharply and robustly with capability, points toward a reframing worth taking seriously: perhaps the empirical signature of “instability” in this data is not primarily about how *often* higher-capability states fight minor conflicts, but about how much *deadlier* their conflicts are once they occur.

A frequency-based reading of the paradox, focused on counts of minor versus major engagements, would have missed this; an intensity-based reading catches it directly, and it is the cleanest, most consistently significant result in the entire analysis.

The mechanism checks behind that finding matter for how broadly it should be read. The battle-death intensity effect is not a global average pulled up by every region equally — it is significantly concentrated in Africa relative to every other region tested, and that concentration survives a full leave-one-out sweep of the African subsample, meaning it is a genuine regional pattern rather than an artefact of one or two heavily weighted countries. At the same time, the effect shows no significant regime-type moderation: it is not a story that is only, or even mostly, about autocracies, nor only about democracies. The combination — broad across regime type, concentrated by region — is a more specific and more falsifiable claim than “capability predicts deadlier conflict everywhere,” and a more useful one for follow-up work asking what, specifically, is different about the conflicts capability is associated with in that region.

RQ2’s null result deserves a precise interpretation, because overclaiming it would be as much a failure of honest reporting as underclaiming it. The finding is that weapon-class acquisitions do not precede measurable shifts in low-intensity conflict activity within a one-to-three-year window, at the level of four broad weapon-category aggregates, tested against a well-calibrated null. That is a real and useful negative result, but it does not, on its own, falsify the “drone effect” hypothesis in the broader sense the original project framing raised it. Absence of evidence for a specific short lead-lag window, at this level of weapon-category aggregation, is not evidence that the underlying mechanism — capability lowering the threshold for low-intensity force projection — does not operate through some other channel: a longer horizon, a finer-grained weapon taxonomy that separates, say, loitering munitions from conventional fixed-wing strike aircraft, or an outcome measure sensitive to conflict types (cyber operations, sub-threshold drone strikes) that this project’s UCDP-based outcome variables do not capture at all. The honest statement is narrower and more useful than either “the drone effect is real” or “the drone effect is false”: within the window and aggregation level tested here, using the most rigorous available test, there is no detectable signal.

The methodological contribution of Section 5.2 is worth treating as a contribution in its own right, independent of what it implies substantively about this project’s data. The Dumitrescu-Hurlin test is a standard, widely used tool in applied panel-causality work across economics and political science, and the within-country shuffle null used to calibrate its significance in textbook implementations is the default choice, not a documented weak point of the method. Any research area that regularly works with panel series exhibiting genuine negative serial correlation — arms-transfer data, procurement cycles, any series that arrives in discrete tranches rather than smoothly — is a candidate for the same miscalibration found here, silently, unless a researcher happens to check the null’s own properties against the real data rather than trusting the standard implementation by default. This paper’s demonstration that a circular-shift null recovers the correct calibration, at comparatively low additional computational cost, is offered as a practical corrective that generalises well beyond this specific dataset.

RQ3’s stability-versus-validity split reinforces a point that follows naturally from treating both diagnostics as answering genuinely different questions rather than as two readings of the same underlying “are these clusters real” question. Bootstrap stability asks whether the same partition would reappear on a different sample; out-of-sample validation asks whether cluster membership carries real forward-looking information regardless of whether the partition itself is perfectly reproducible. A partition can fail the first test for two of its four categories and still pass the second test unanimously across all four — which is exactly the pattern found here — and the correct response is not to treat one test as trumping the other, but to report both and let the reader judge what each is actually evidence for. Practically,

this means the Safe Hegemon and Inert Non-Combatant archetypes should be read as comparatively durable categories, while Defensive Deterrent and Armed Instabilizer are better understood as real, predictively useful, but softer regions on a continuum rather than sharply bounded types.

Read across all three research questions, a single thread connects results that on the surface look unrelated: nearly every strong claim this project could have made on a first pass turned out to need qualification once tested properly, and in every case the qualification made the finding more precise rather than simply weaker. RQ1's amplification finding is sharper than a vague "capability correlates with conflict" claim precisely because the substitution hypothesis was tested directly and rejected, leaving intensity rather than composition as the actual signature worth explaining. RQ2's null is more informative than a single non-significant p-value because it survives two independent tests built on different logics, and because the process of getting there identified a general problem with how this class of test is usually run. RQ3's archetypes are more credible specifically because half of them failed a properly calibrated stability check — a paper reporting four uniformly "stable" clusters under the wrong threshold would have been reporting an artefact of the threshold, not a finding about the world. None of these refinements were available from the first pass of any analysis in this project; all three came from deliberately testing the first pass against something more rigorous before trusting it.

7. Limitations

Several limitations bound how far these results should be read. First, endogeneity is not resolved by this design. High-capability states — Israel, South Korea, and Taiwan are the clearest examples — often hold that capability *because* of pre-existing security threats, not the other way around; the two-way fixed-effects specification absorbs time-invariant country traits and global time trends, which addresses some of the resulting confounding, but it does not eliminate reverse causality, and the results throughout this paper are reported as associational rather than causal for that reason. Second, the primary conflict data source, UCDP, applies a 25-battle-death-per-year floor to what counts as organised armed violence, and does not capture cyber operations or sub-threshold drone strikes that stay below that floor — activity that is, by the theoretical framing motivating this project, plausibly part of the "instability" mechanism itself, meaning the outcome variables used here may systematically undercount the very phenomenon the paradox predicts. Third, the effective analysis window is 1989–2024 rather than the full nominal 1946–2024 range that some source datasets notionally cover; results say nothing directly about the Cold War decades before 1989, where UCDP's finer-grained event data, V-Dem's coverage, and SIPRI's deal-level transfer data are all considerably less complete. Fourth, and specific to RQ2, the null result is bounded by design choices that were reasonable but not the only reasonable ones: a lag window of one to three years, and four broad weapon-class aggregates rather than a finer-grained taxonomy. The null reported here does not rule out lead-lag relationships operating over longer horizons or at a more disaggregated weapon-category level than this project's four-class grouping was built to detect. Fifth, the RQ3 clustering results should be read with an eye to the fact that K-Means cluster identities are not automatically stable across independent model refits: the out-of-sample validation exercise in Section 5.3 uses a separately trained clustering on pre-2019 data, whose numeric cluster labels do not correspond one-to-one with the main-sample archetype labels by construction, and the interpretation offered there — that the out-of-sample groups predict future conflict — deliberately does not lean on any claimed correspondence between the two cluster labelings beyond what the predictive test itself supports. Finally, the mechanism checks reported in Section 5.1 and Section 6 (the regional concentration and regime-type interaction results) were run as targeted robustness probes on a finding that had already emerged from the primary specification, not as inde-

pendently pre-registered tests; they should be read as consistent with the primary result rather than as fully independent confirmatory evidence of it.

8. Conclusion

This paper set out to test whether the stability-instability paradox, developed to describe nuclear deterrence, generalises to the conventional precision-strike capability that has reshaped force projection over the past two decades. The answer is qualified rather than clean, and reported that way throughout: capability does not buy states a substitution from war toward minor conflict, as the classical compositional reading of the paradox would predict, but it is associated with more conflict participation of both kinds and, most sharply, with deadlier conflict once it occurs — an amplification story rather than a substitution story, and arguably a more consequential one. The weapon-class lead-lag mechanism that would have supplied a proximate, testable link between specific capability and near-term instability does not survive a properly calibrated test, a result reached independently through inference and through genuine out-of-sample forecasting, and the process of discovering *why* the naive version of that test looked promising is offered here as a methodological finding with relevance well beyond this project’s own data. Countries do sort into recognisable, forward-looking conflict-capability archetypes, but only half of those archetypes meet a rigorously calibrated bar for statistical stability, a result this paper reports as a genuine split rather than smoothing in either direction.

Every one of these conclusions rests on a result that was run through a falsification, permutation, or out-of-sample test before being trusted — and in at least one case, that test revealed something about the *method* being used to answer the question, not just about the answer itself. That is offered here as the project’s central methodological lesson: a result that has not been checked against a properly specified null is not yet a result. Future work extending this design could usefully push in three directions this project’s scope did not allow: a finer-grained weapon-category taxonomy for RQ2, capable of distinguishing loitering munitions and other genuinely novel systems from the broader ground- and air-systems categories used here; conflict outcome measures that reach below UCDP’s battle-death floor into the cyber and sub-threshold activity this paper’s Limitations section flags as a coverage gap; and a longer lead-lag horizon than the one-to-three-year window tested here, since a mechanism operating through slower channels — training, doctrine adaptation, institutional learning — would not have been visible to any of the three tests run in Section 5.2. None of these extensions would overturn the results reported here; each would test whether the null found at this window and this level of aggregation is a genuine absence of the mechanism, or a boundary of what this particular design could detect — a distinction this paper has tried to keep visible throughout rather than collapsing into a single, simpler-sounding conclusion.

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Data Availability / Replication Statement

All analysis code is available in the project's GitHub repository. Raw data are not redistributed in the repository, in keeping with the licensing terms of each source dataset; each source's download location and license terms are documented in the repository's data/README.md, and the analysis pipeline expects raw files to be placed into the directory layout described there before the notebooks are run. Notebooks are numbered for sequential execution order from data acquisition through final synthesis; running them in numeric order reproduces every table and figure cited in this paper from the same raw inputs.

Author Statement

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